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elfe3D: Modelling with the total electric field approach using finite elements in 3D

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Summary

The controlled-source electromagnetic method is a geophysical technique that detects variations in electrical and magnetic material properties in the subsurface. The method is valuable for both environmental assessments and resource exploration. To evaluate electromagnetic data obtained over complex subsurface structures, precise three-dimensional numerical modeling software is required. This is particularly important for tasks related to survey design and for inversion routines that estimate subsurface models based on field data inputs. The forward modelling program `elfe3D` simulates electric and magnetic field responses for three-dimensional subsurface models, i.e. the data of frequency-domain controlled-source electromagnetic configurations. To optimise the balance between the size of the problems and the accuracy of the solutions, adaptive mesh refinement strategies are employed. The `elfe3D` program allows the user to define an arbitrary survey setup: The sensor locations, where the electric and magnetic field responses are calculated, can be placed at or below the subsurface, or in the air. This flexibility facilitates realistic modelling of surface-based, borehole and airborne controlled-source electromagnetic surveys.

Statement of need

The numerical modelling program `elfe3D` calculates forward responses for frequency-domain three-dimensional (3D) controlled-source electromagnetic surveys in geophysics. The physical behavior of electromagnetic fields is described by Maxwell's equations, which serve as the foundation for the curl-curl equation in terms of the total electric field $\mathbf{E}$ that `elfe3D` is solving:

$$\nabla \times \frac{1}{\mu} \nabla \times \mathbf{E} - i\omega \frac{1}{\rho} \mathbf{E} - \omega^2 \epsilon \mathbf{E} = i\omega \mathbf{J}_p \quad \text{in} \quad \Omega,$$

where Ω is the computational domain, ω is the angular frequency, ρ the electrical resistivity, ε the dielectric permittivity and μ the magnetic permeability. A time dependence $e^{-i\omega t}$ is assumed and $\mathbf{J}_p$ describes the source term. Dirichlet boundary conditions are imposed on the outer domain boundaries.

Isotropic electric resistivities and magnetic permeabilities are variable, but element-wise constant, model parameters arranged in a tetrahedral mesh generated with `TetGen` (Si, 2015). Line or loop sources are represented along the edges of the mesh elements. The governing equations are discretised with linear vector finite-element shape functions (Jin, 2014). Developed using modern Fortran, `elfe3D` leverages vectorisation and shared-memory parallelism and employs a direct method, the `MUMPS` solver (Amestoy et al., 2001), for solving the system of equations.

Standard strategies in electromagnetic modelling often avoid 3D simulations due to their substantial computational demands and the predominant focus tends to be on electrical

resistivity models, as this material property is dominant in the curl-curl equation at typical frequencies used in controlled-source electromagnetic surveys. Contemporary controlled-source electromagnetic field investigations employ advanced measurement configurations designed to image complex subsurface structures. This context necessitates the development of flexible and manageable 3D simulation software tailored for the evaluation of controlled-source electromagnetic data. The numerical techniques required to facilitate this advancement are sophisticated, and their integration into controlled-source electromagnetic modelling code is currently a prominent area of academic research (Castillo-Reyes et al., 2018; Grayver & Kolev, 2015; Heagy et al., 2017; Rochlitz et al., 2019; Werthmüller et al., 2019).

In `elfe3D`, three state-of-the-art functionalities are combined:

1. **Variable Magnetic Permeability**: In traditional 3D electromagnetic forward modeling algorithms, magnetic permeability and dielectric permittivity are typically not considered variable model parameters. This oversight is often due to the relatively minor contrasts in these properties compared to conductivity contrasts prevalent in most geological environments at frequencies in the Hz- to kHz-range. However, it is well documented that mineralised zones and metallic infrastructure (Heagy & Oldenburg, 2023) may exhibit pronounced contrasts in magnetic permeability relative to their host rocks. Dielectric permittivities influence only high-frequency data in very resistive environments (Kalscheuer et al., 2008). Consequently, in `elfe3D`, magnetic permeability is treated as a variable model parameter, in conjunction with electrical resistivity.

2. **Automatic Mesh Refinement**: The refinement approach implemented in `elfe3D` is inspired by Ren et al. (2013). It aims to achieve an optimal balance between the accuracy of the solutions at specified areas of interest and the overall size of the computational problem. This is accomplished through a goal-oriented mesh refinement strategy, specifically tailored for models characterised by variable electrical conductivity and magnetic permeability.

3. **Perfect Electric Conductors**: `elfe3D` incorporates a methodology for approximating highly conductive infrastructures within the model, such as steel-cased wells that are frequently encountered in exploration contexts. These structures are modeled as perfect electric conductors (Um et al., 2020), utilising an approach similar to boundary conditions. Within the computational domain, this formulation enforces the electric field to zero at mesh edges corresponding to the perfect electric conductors.

The program `elfe3D` is designed for geophysicists who want to:

- Plan controlled-source electromagnetic experiments and require a tool for conducting numerical feasibility studies.
- Generate problem-specific tetrahedral meshes for controlled-source electromagnetic settings, which can be utilised in other simulation environments.
- Integrate the code into their inversion frameworks.
- Further develop the code.

The underlying code of `elfe3D` was developed between 2018-2023 and validated in Rulff et al. (2021). This initial code is implemented in the inversion software `emilia` (Kalscheuer et al., 2008, 2010, 2015) to enable 3D controlled-source electromagnetic inversion (Rulff, 2023; Rulff & Kalscheuer, 2024). Adaptively refined meshes and parts of the synthetic data reported in Castillo-Reyes et al. (2023) were generated with `elfe3D`. The program is currently used to design surface-to-borehole controlled-source electromagnetic surveys for geothermal applications (Rulff et al., 2024).

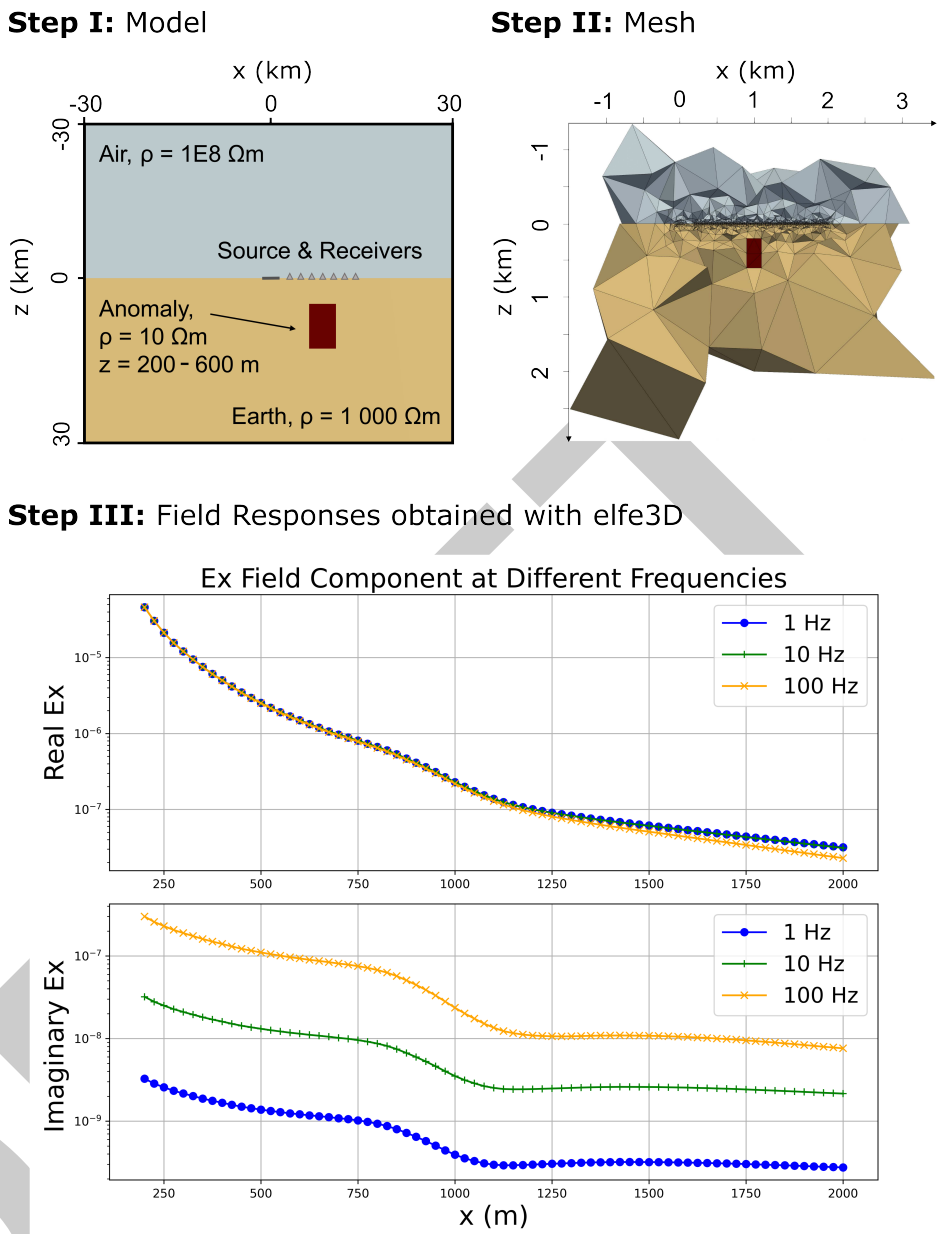

Figure 1: Key steps of the forward modelling procedure including the choice of a subsurface model and source-receiver setup (Step I; note that the subsurface anomaly and survey setup are enlarged in the image for better visibility), the meshing of the modelling domain (Step II; note that a slice through the inner modelling domain is displayed) and the calculation of field responses with `elfe3D` (Step III; note that only the Ex field component is displayed). Figure adapted from PhD thesis: Rulff (2023).

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